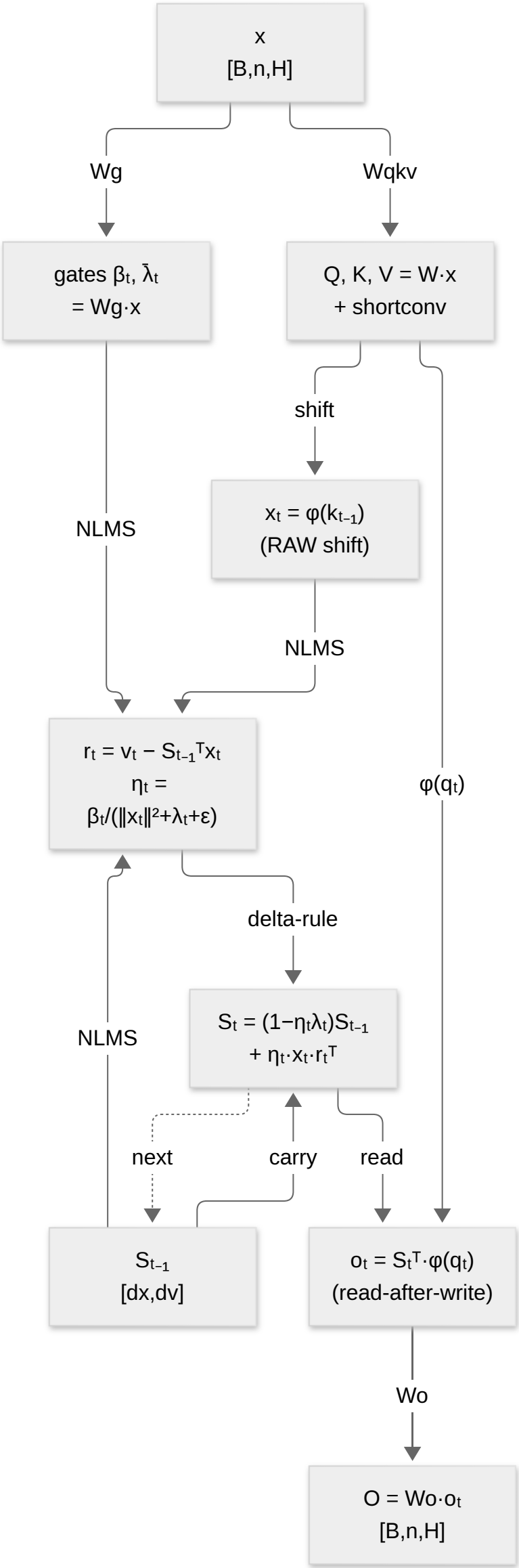

x
[B,n,H]
Wg
Wqkv
gates β_t, $\bar{\lambda}_t$
= Wg·x
Q, K, V = W·x
+ shortconv
shift
NLMS
$x_t = \varphi(k_{t-1})$
(RAW shift)
NLMS
$r_t = v_t - S_{t-1}^{\top} x_t$
$\eta_t =$
$\beta_t/(\|x_t\|^2+\lambda_t+\varepsilon)$
$\varphi(q_t)$
delta-rule
NLMS
$S_t = (1-\eta_t\lambda_t)S_{t-1}$
$+ \eta_t \cdot x_t \cdot r_t^{\top}$
next
carry
read
S_{t-1}
[dx,dv]
$o_t = S_t^{\top} \cdot \varphi(q_t)$
(read-after-write)
Wo
O = Wo·o_t
[B,n,H]